Université Abdelmalek Essaâdi

L'Ecole Nationale des Sciences Appliquées d'Al-Hoceima

Filière :Transformation Digitale et Intelligence Artificielle

Automatic speech recognition for Moroccan Darija and Tarifit:

Optimizing OpenAI Whisper for a resource-constrained dialect

Réalisé par :

EL-KHAMLICHI NADA

ESSAIDY NABILA

Encadré par :

Pr.M.CHERRADI

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2. Research Paper

2.1. Title

Automatic speech recognition for Moroccan Darija and Tarifit: Optimizing OpenAI Whisper for a resource-constrained dialect.

2.2. Authors

EL-khamlichi Nada

Essaidy Nabila

Supervised by: Mohammed Cherradi

2.3. Abstract

Automatic Speech Recognition (ASR) systems have achieved remarkable performance on high-resource languages, yet their effectiveness remains limited for under-resourced dialects and minority languages such as Moroccan Darija and Tarifit. This work presents a comparative study on the adaptation of OpenAI Whisper models for both Moroccan Darija and Tarifit in resource-constrained environments.

For Darija, several Whisper variants were evaluated before fine-tuning Whisper Small on a subset of approximately 4,500 training samples extracted from a publicly available Moroccan Arabic speech corpus. The baseline evaluation produced a Word Error Rate (WER) of 133.4%, highlighting the limitations of the pre-trained model on Moroccan dialectal speech. After fine-tuning, the WER decreased to 45.8%, corresponding to a relative improvement of 65.7%.

For Tarifit, the lack of publicly available annotated corpora required the manual creation of a dedicated dataset. A speech corpus containing 121 audio samples was collected from YouTube sources, converted into WAV format, and manually transcribed using Latin script. The dataset was divided into training, validation, and testing subsets. Five Whisper variants — Tiny, Base, Small, Medium, and Large-v3 — were then evaluated on Tarifit speech. The baseline experiments produced very high error rates, with Whisper Large-v3 achieving the best initial performance at 121.67% WER. Whisper Small was selected for fine-tuning due to its balance between performance and computational feasibility on available hardware resources. After fine-tuning, the WER improved from 114.58% to 88.75%, representing a relative improvement of 22.55%.

The experiments were conducted using the Hugging Face Transformers framework on NVIDIA T4 GPUs within a Kaggle environment. In addition, visualization tools and interactive notebook-based interfaces were developed to compare baseline and fine-tuned transcription outputs.

The results demonstrate that targeted fine-tuning, even on limited datasets, can substantially improve ASR performance for low-resource Moroccan languages. This work also highlights the importance of dataset creation, transfer learning, and accessible ASR research for dialectal Arabic and Amazigh languages.

2.4. Keywords:

Automatic Speech Recognition; Moroccan Darija; Whisper; Fine-tuning; Word Error Rate; Low-resource languages; Arabic dialects; Transfer learning; Huggingt Face; Deep learning.

2.5. Introduction

2.5.1. Context

In recent years, deep learning-based Automatic Speech Recognition (ASR) systems have achieved near-human performance on several high-resource languages, including English, French, and Mandarin. OpenAI's Whisper model [1], trained on 680,000 hours of multilingual audio data, represents one of the most capable general-purpose ASR systems available to date. However, the linguistic diversity of the world extends far beyond the languages well-represented in current training corpora. Morocco alone is home to multiple spoken varieties, including Modern Standard Arabic (MSA), Moroccan Arabic — commonly known as Darija — Tamazight, and Tarifit. Of these, Darija is the most widely spoken in everyday communication, yet it remains one of the least supported dialects in state-of-the-art ASR research. In addition, Tarifit — a major Amazigh language spoken primarily in Northern Morocco — suffers from an even more severe lack of annotated speech resources, benchmarks, and dedicated ASR studies, despite its strong cultural and linguistic importance.

2.5.2. Problem Statement

Darija presents several linguistic challenges that make speech recognition particularly difficult. Unlike standardized languages, Darija varies significantly from one region to another, whether in Fes, Casablanca, Oujda, or Al Hoceima. It also incorporates many words and expressions borrowed from French, Spanish, and Amazigh, which increases its linguistic complexity. Another major challenge is the absence of a standardized writing system, as the same sentence can be written in multiple ways depending on the speaker or context. In addition, Darija is still largely underrepresented in the large speech datasets used to train modern ASR systems. Because of these limitations, directly applying Whisper to Moroccan Arabic often leads to very high transcription errors, making the system unreliable for practical use. This limitation affects the development of applications such as voice assistants, accessibility technologies, and automatic transcription services designed for Moroccan users.

2.5.3. Contributions

This work makes several important contributions to the study of Automatic Speech Recognition for under-resourced Moroccan languages, particularly Moroccan Darija and Tarifit.

First, we conduct a comprehensive benchmark evaluation of multiple OpenAI Whisper variants — Tiny, Base, Small, Medium, and Large-v3 — on both Darija and Tarifit speech data. This systematic comparison highlights the limitations of pre-trained multilingual ASR systems when applied directly to Moroccan low-resource languages.

For Darija, all evaluated models produced high error rates, with Word Error Rates exceeding 85%, while Whisper Large achieved the best baseline performance with a WER of 85.3%.

For Tarifit, the evaluation also revealed severe performance limitations, with WER values ranging from 121.67% for Whisper Large-v3 to 165.42% for Whisper Base. Additional baseline results for Tarifit include 129.58% WER for Whisper Tiny, 152.92% for Whisper Small, and 132.50% for Whisper Medium. These results confirm that Whisper models struggle significantly with Moroccan dialectal and Amazigh speech without language-specific adaptation.

Second, this work contributes to the development of speech resources for underrepresented Moroccan languages. While the Darija experiments relied on a publicly available speech corpus, no suitable annotated dataset was available for Tarifit. To address this limitation, a dedicated Tarifit dataset was manually created for this project. The corpus contains 121 audio samples collected from YouTube sources, converted into WAV format, and manually transcribed using Latin script. The dataset was subsequently divided into training, validation, and testing subsets to support reproducible experimentation.

Third, based on the benchmark evaluation and the computational limitations of the available hardware environment, Whisper Small was selected as the most suitable model for fine-tuning on both languages. Although larger models such as Whisper Large-v3 achieved slightly better baseline performance, their computational requirements made them impractical for fine-tuning on NVIDIA T4 GPUs within a Kaggle-based environment. Whisper Small therefore represented the best compromise between transcription quality, accessibility, and training feasibility.

Fourth, we establish clear performance baselines and demonstrate substantial improvements after fine-tuning. For Darija, Whisper Small initially produced a WER of 133.4%. After fine-tuning on approximately 4,500 curated training samples, the WER decreased to 45.8%, corresponding to a relative improvement of 65.7%. For Tarifit, the baseline evaluation of Whisper Small produced a WER of 114.58% and a CER of 84.74%. After fine-tuning on the custom Tarifit dataset, the WER improved to 88.75% and the CER to 84.36%, corresponding to a relative WER improvement of 22.55%.

Fifth, to support analysis and reproducibility, the project includes the generation of multiple evaluation visualizations and comparative performance graphs illustrating training evolution, WER progression, and differences between baseline and fine-tuned models. In addition, the fine-tuned Darija model was made publicly available on the Hugging Face Hub under the identifier `maghreb-ai-research/whisper-small-darija`, accompanied by an interactive comparison interface allowing users to evaluate transcription outputs in real time.

Finally, this work contributes to ongoing efforts toward more inclusive ASR technologies for Moroccan dialects and Amazigh languages. By combining dataset creation, transfer learning, model evaluation, and practical experimentation in resource-constrained environments, the project establishes a foundation for future research involving larger datasets, advanced Whisper architectures, and broader Moroccan multilingual ASR systems.

2.5.4. Paper Organization

The rest of this paper is structured as follows. In Section 2.6, we review existing work on speech recognition for low-resource languages and Arabic dialects, paying particular attention to studies that have explored Whisper and related sequence-to-sequence architectures in similar settings. Section 2.7 then walks through our methodology, covering the dataset we worked with, how we preprocessed the audio data, why we selected Whisper Small for fine-tuning, and the details of our training setup. In Section 2.8, we report and discuss our experimental results, from the initial benchmark comparing all Whisper variants to the training dynamics and final evaluation of our fine-tuned model. Finally, Section 2.9 closes the paper with a reflection on what we achieved, an honest look at the limitations we encountered, and suggestions for where this work could go next.

2.6. Literature Review

2.6.1. ASR for Arabic and Its Dialects

Arabic is a linguistically rich language with a wide variety of spoken dialects that can differ quite dramatically from Modern Standard Arabic in terms of vocabulary, pronunciation, and sentence structure. For a long time, ASR research on Arabic concentrated almost exclusively on MSA, where systems built around Hidden Markov Models combined with n-gram language models managed to deliver solid results [2]. Extending those systems to dialectal Arabic, however, turned out to be a much harder problem — partly because dialects have no standardized writing conventions, and partly because labeled speech data for them has always been scarce.

As deep learning gradually took over the field, researchers started tackling dialectal Arabic with more flexible architectures. Khurana and Ali [3] explored a hybrid CNN-RNN setup for Egyptian Arabic, while Bougrine et al. [4] ran a series of experiments on Algerian dialects and found that performance still lagged noticeably behind what was achievable on MSA. Closer to our own focus, Boujraf et al. [5] put together a small Darija speech corpus and tested Kaldi-based acoustic models on it, ending up with WER values above 60%. What none of these works had done, though, was ask whether a large multilingual model like Whisper — fine-tuned rather than trained from scratch — could change the picture for Moroccan Arabic.

2.6.2. Transfer Learning and Fine-Tuning for ASR

The idea of fine-tuning large pre-trained models rather than building everything from the ground up has become central to modern ASR. Models like wav2vec 2.0 [6], HuBERT [7], and Whisper [1] have shown a remarkable ability to transfer across languages and acoustic conditions. Whisper in particular was trained by Radford et al. [1] on a massive and diverse web-scraped dataset in a weakly supervised fashion, which gave it surprisingly broad linguistic coverage and decent noise robustness out of the box. Several teams have since shown that fine-tuning it on a specific low-resource language can yield dramatic improvements: Peng et al. [8] cut the WER by more than half on Cantonese, and Nowakowski et al. [9] reported comparable gains for Polish.

Especially relevant to what we are doing here, Daouad et al. [10] investigated the fine-tuning of Whisper for Tarifit — a dialect of Amazigh spoken in Morocco that, much like Darija, is a low-resource language with very little annotated data to work with. Their study explored parameter-efficient adaptation techniques including LoRA, DoRA, AdaLoRA, and QLoRA, and demonstrated that competitive ASR performance can be reached even when only around 6% of the model's parameters are actually updated during training. The linguistic proximity of Tarifit to Darija, combined with the methodological similarities between their setup and ours, made their findings particularly informative when designing our own approach.

2.6.3. Positioning of This Work

As far as we know, no prior study has carried out a full cross-model benchmark of all Whisper variants on Moroccan Darija and then gone on to fine-tune the most practical candidate on a publicly available corpus. Beyond the technical results, we also built an interactive Gradio interface so that anyone can directly hear the difference between the original and fine-tuned model — which we believe makes the contribution more concrete and accessible than a purely offline evaluation would allow.

2.7. Methodology

2.7.1. Structure and Headings

Our methodology follows a structured pipeline that moves from data preparation to model evaluation, with a deliberate model selection step in between. The overall goal is to adapt a large pre-trained multilingual ASR model — specifically OpenAI's Whisper — to the specific acoustic and linguistic characteristics of Moroccan Darija, a dialect that is virtually absent from the model's original training distribution. The pipeline consists of four main stages: dataset preparation, baseline benchmarking across all Whisper variants, fine-tuning of the selected model, and final evaluation.

2.7.2. Notation

2.7.3. Theoretical Framework

Whisper is a sequence-to-sequence model based on the Transformer architecture. It takes log-Mel spectrograms as input and produces text tokens as output through an encoder-decoder structure. The model was pre-trained by Radford et al. [1] on approximately 680,000 hours of multilingual audio in a weakly supervised fashion, making it one of the most broadly capable ASR models available.

Fine-tuning in this context follows the standard transfer learning paradigm: rather than training a model from random initialization, we start from the pre-trained weights of Whisper Small and continue training on our Darija-specific dataset. The intuition is that the encoder has already learned general acoustic representations — phoneme-level patterns, prosody, noise robustness — and what is needed is to adapt the decoder's language model component to the specific vocabulary and pronunciation patterns of Darija. The model is optimized by minimizing the cross-entropy loss between the predicted token sequence and the reference transcription:

$$\mathcal{L} = -\sum_{t-1}^{T} \log P(y_t \mid y_{<t}, x)$$

where y_t is the ground-truth token at position tt t and x is the encoded audio representation.

2.7.4. Method Description

Dataset :

This study relies on two distinct datasets corresponding to the two target languages: Moroccan Darija and Tarifit.

For Darija, we used the publicly available dataset ***afyfbadreddine77/darija-asr-dataset***, hosted on the Hugging Face Hub. The corpus consists of approximately 12,200 audio samples in WAV format, with durations ranging from 0.44 to 27.4 seconds, each paired with a transcription in Arabic script. The transcriptions vary in length from 4 to 410 characters, reflecting a wide range of utterance types — from very short responses to longer conversational sentences. All recordings are in Moroccan Darija as it is naturally spoken, meaning that the language is colloquial, often code-switched with French, and phonetically variable across speakers.

Due to the computational constraints of the local training environment — particularly limited GPU memory and processing capacity — it was not feasible to fine-tune the model on the entire 12,200-sample corpus. Therefore, a subset of approximately 5,000 samples was selected for experimentation and fine-tuning, representing a practical upper bound given the available hardware resources. This subset was divided into a training set (90%, approximately 4,500 samples) and a validation set (10%, approximately 500 samples), using a fixed random seed of 42 to ensure reproducibility.

While this limitation restricts the maximum achievable performance of the fine-tuned model, it also reflects a realistic scenario commonly encountered in low-resource and resource-constrained ASR research settings.

For Tarifit, no suitable public ASR dataset was available. Consequently, a dedicated speech corpus was manually created specifically for this project. Audio samples were collected from publicly available YouTube videos featuring native Tarifit speakers. The videos were converted into WAV audio format, segmented into smaller utterances(from 10 to 20 seconds), and manually transcribed using Latin-script transcription conventions.

The final Tarifit dataset contains 121 audio samples. Given the relatively small size of the dataset, careful train, validation, and test splits were applied in order to maximize training efficiency while preserving evaluation reliability.

More specifically, 90 audio samples were allocated to the training set for fine-tuning, while 15 samples were reserved for evaluation before fine-tuning and an additional 15 samples were used for evaluation after fine-tuning in order to compare model performance across both stages. The scarcity of Tarifit speech resources highlights one of the major challenges in low-resource ASR research, as Amazigh language varieties still suffer from a severe lack of annotated speech corpora.

For both datasets, all audio files were standardized to a sampling rate of 16,000 Hz, corresponding to Whisper's required input format. In addition, a text normalization procedure was applied to all transcriptions prior to training and evaluation. This process included removing punctuation marks and collapsing multiple whitespace characters to ensure consistency during Word Error Rate (WER) computation.

The significant difference in dataset scale between Darija and Tarifit further illustrates the imbalance in available speech resources: while dialectal Arabic corpora are gradually becoming publicly accessible, Amazigh varieties such as Tarifit remain significantly underrepresented in existing ASR research and datasets.

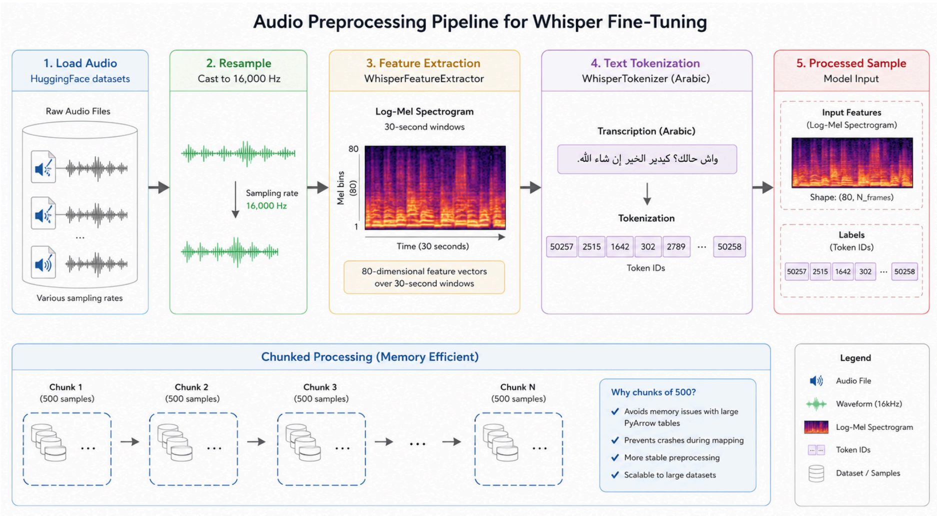
Model Selection :

Before committing to any single model for fine-tuning, we ran a systematic benchmark across all five Whisper variants — Tiny, Base, Small, Medium, and Large — on a held-out subset of 10 audio samples. Each model was evaluated in zero-shot mode with the language forced to Arabic, and WER was computed for each sample. The results, summarized in Table 1, show that all models perform poorly on Darija out of the box, with WER values ranging from 85.3% (Large) to 101.7% (Tiny). While the Large model achieves the best zero-shot performance, it requires substantial GPU memory and cannot be fine-tuned on a standard local machine. Whisper Small, which ranks second with a WER of 91.3%, represents the best practical trade-off between performance and computational accessibility, and was therefore selected for fine-tuning.

```
======================
RESULTATS FINAUX
======================
tiny ---> 1.017
base ---> 0.953
small ---> 0.913
medium ---> 0.928
large ---> 0.853
```

Preprocessing Pipeline : #khesso ytbdel

Audio preprocessing was handled using the HuggingFace `datasets` library. Each audio file was loaded and cast to a sampling rate of 16,000 Hz. Log-Mel spectrograms were then extracted using Whisper's built-in `WhisperFeatureExtractor`, producing 80-dimensional feature vectors over 30-second windows. On the text side, transcriptions were tokenized using the `WhisperTokenizer` configured for Arabic. To handle memory limitations associated with large PyArrow tables, the preprocessing was applied in chunks of 500 samples rather than all at once.



2.7.5. Experimental Protocol

Fine-tuning was performed using the Hugging Face `Seq2SeqTrainer` framework with mixed-precision training (`fp16=True`) enabled to reduce GPU memory consumption and speed up training. The model was initialized from `openai/whisper-small`, while the language setting was configured for Arabic transcription. To give the model greater flexibility in learning Moroccan Darija patterns, parameters such as `forced_decoder_ids`, `suppress_tokens`, and `begin_suppress_tokens` were disabled, allowing the decoder to generate transcriptions more naturally without predefined constraints.

The training setup used a learning rate of (1 \times 10^{-5}) with a linear warmup over the first 200 steps. A batch size of 2 samples per GPU was used, combined with gradient accumulation over 2 steps, resulting in an effective batch size of 4 across two GPUs. Training was scheduled for a maximum of 10 epochs, with evaluation and checkpoint saving performed after each epoch. Model selection was based on the lowest Word Error Rate (WER) obtained on the validation set. In addition, early stopping was applied to automatically stop training when no further improvement was observed for three consecutive epochs. Gradient checkpointing was also enabled to optimize GPU memory usage during backpropagation.

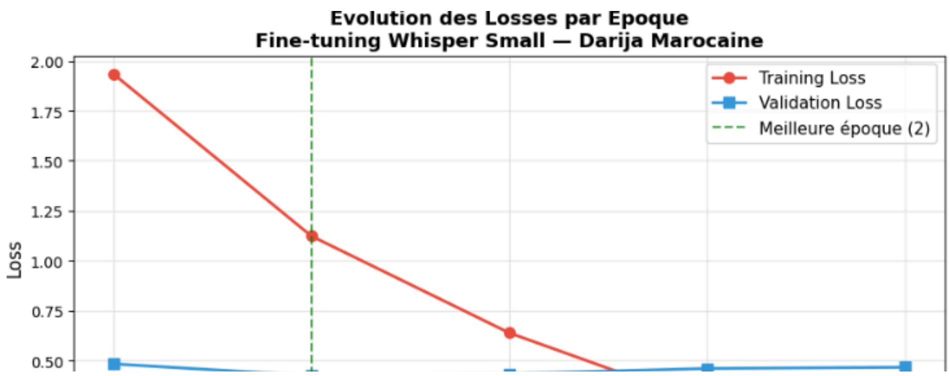
Before starting the fine-tuning process, an initial evaluation was conducted on 50 validation samples using the original pre-trained Whisper Small model. The experiment produced a WER of 133.4%, highlighting the model's difficulty in accurately transcribing Moroccan Darija in its default state. This baseline served as the main reference point for evaluating the effectiveness of the fine-tuning process.

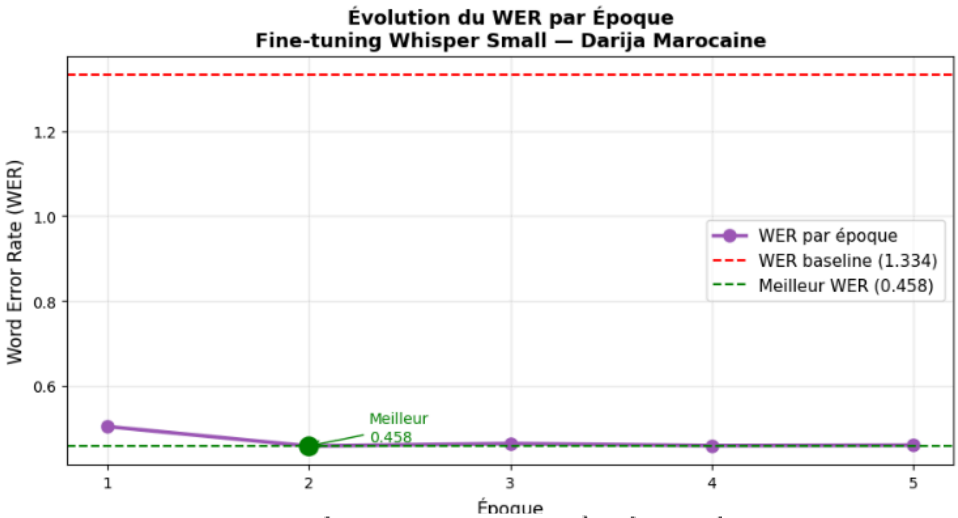
Training ultimately stopped after 5 epochs because early stopping criteria were met, while the best-performing checkpoint was obtained at epoch 2. The final fine-tuned model was then saved locally and published on the Hugging Face Hub under the identifier "maghreb-ai-research/whisper-small-darija" to support reproducibility and future research on Moroccan dialect ASR systems.

2.8. Results

2.8.1. Tables and Figures

Fine-tuning of Darija:





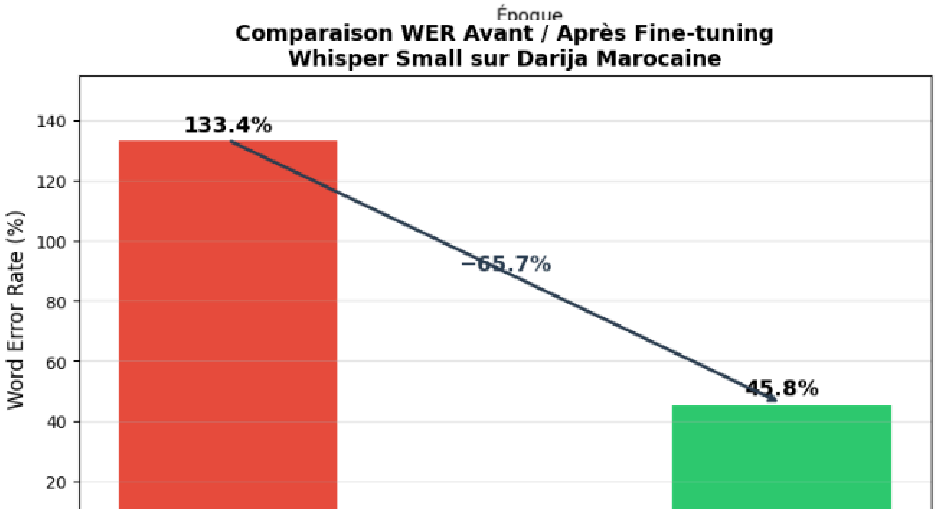
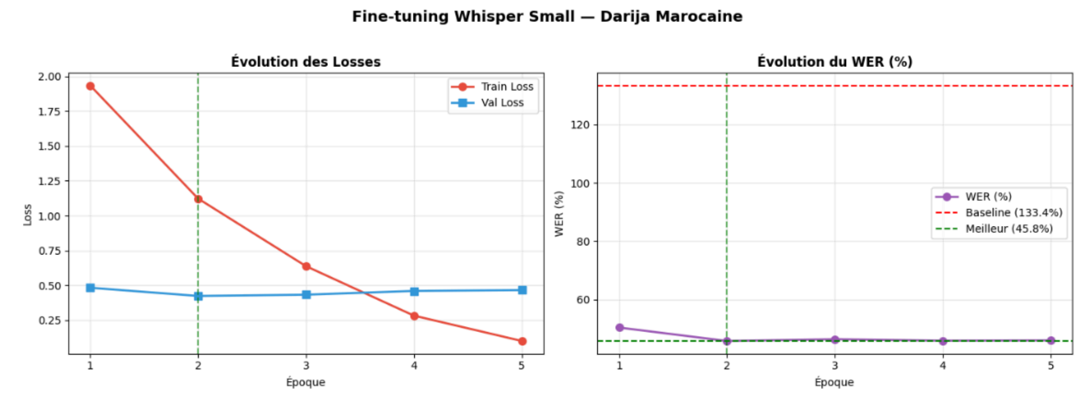
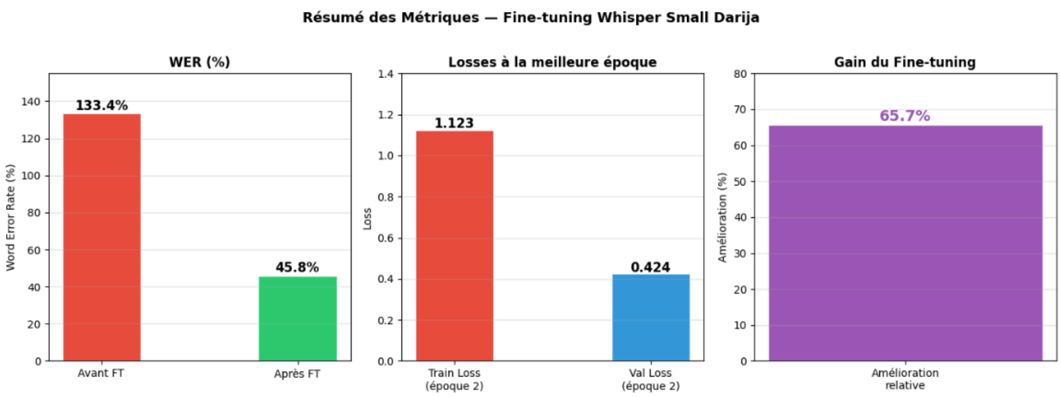


Tableau Récapitulatif des Métriques
Fine-tuning Whisper Small — Darija Marocaine

Métrique	Avant FT	Après FT	Gain	Tendance
WER (Word Error Rate)	133.4%	45.8%	−65.7%	↓ Meilleur
Meilleure époque	—	Époque 2	—	—
Training Loss (finale)	—	0.101	—	↓ Meilleur
Validation Loss (meilleure)	—	0.424	—	↓ Meilleur
Nb échantillons entraînemen	—	4 500	—	—
Nb échantillons test	—	500	—	—
Modèle de base	Whisper Small	Whisper Small	—	—





2.8.2. Statistical Analysis

Fine-tuning of Darija:

This section walks through the main numerical results of our fine-tuning experiments, building on the plots presented in Section 2.8.1.

Baseline Performance. Before any fine-tuning, we evaluated the original Whisper Small model on 50 Darija validation samples and obtained a WER of 133.4%. A value above 100% means the model was not just making transcription mistakes — it was actively generating words that had nothing to do with what was actually said. This kind of hallucination behavior is typical when a model is applied to a language variety it has never meaningfully encountered during training, and it made clear from the outset that some form of adaptation would be necessary.

Training Dynamics. Looking at how the losses evolved over training, the picture is fairly straightforward. The training loss dropped steadily from around 1.97 at epoch 1 down to 0.15 by epoch 5, which shows the model was learning. The validation loss, on the other hand, hit its lowest point at epoch 2 (around 0.42) and barely moved after that. This is what ultimately triggered early stopping — after three epochs with no meaningful gain on the validation set, training was automatically halted.

WER Progression. The WER curve tells much the same story. The model reached its best performance at epoch 2, with a WER of 45.8%, and this checkpoint was saved as the final model. After that point, even as the training loss kept falling, the validation WER showed no further improvement — a sign that the model had started fitting the training examples too closely rather than generalizing.

Before/After Comparison. Putting the two numbers side by side, the fine-tuned model reduced the WER from 133.4% to 45.8% — an absolute drop of 87.6 percentage points, or a relative improvement of 65.7%. Given that this was achieved with only around 4,500 training samples, we found this result encouraging, and it suggests that even modest amounts of targeted data can make a real difference for an underrepresented dialect.

Cross-Model Benchmark. Across the five Whisper variants tested in zero-shot mode, WER ranged from 101.7% for Tiny down to 85.3% for Large, with Small landing at 91.3%. What this tells us is that simply using a bigger model does not solve the problem — all variants struggled considerably with Darija out of the box. Fine-tuning, rather than scale, turned out to be the deciding factor.

2.8.3. Interpretation of Results

Fine-tuning of Darija:

The results we obtained raise a few natural questions — why did the model struggle so much at the start, why did fine-tuning help as much as it did given how little data we had, and why did performance stop improving after epoch 2? This section tries to answer each of these in turn.

Why the baseline performs so poorly. A WER of 133.4% is striking, but it makes sense once you consider what Whisper was actually trained on. Although Arabic is part of its training data, the Arabic it has seen is overwhelmingly Modern Standard Arabic — the formal, written variety used in news broadcasts and official contexts. Moroccan Darija is something else entirely. It has its own vocabulary, its own pronunciation patterns, and it regularly pulls in words from French, Spanish, and Amazigh in the middle of otherwise Arabic sentences. When Whisper encounters Darija speech, it has no real frame of reference for it and ends up mapping what it hears onto MSA-like output, producing transcriptions that often bear little resemblance to what was actually said. The hallucinated words — those that appear in the model's output but not in the reference — are a direct consequence of this mismatch.

Why fine-tuning works even with limited data. The 65.7% relative improvement we achieved with roughly 4,500 samples is a good illustration of why transfer learning matters in low-resource settings. We were not asking the model to learn speech recognition from scratch — Whisper's encoder had already developed strong general-purpose acoustic representations during pre-training, capturing things like phoneme boundaries, prosody, and speaker variation. What fine-tuning did was essentially redirect the decoder: instead of generating MSA-style output, it gradually learned to produce text that matches the vocabulary and spelling patterns found in our Darija transcriptions. That is a much more tractable problem than building an ASR system from zero, and it explains why a relatively small dataset was enough to move the needle significantly.

Why performance plateaued after epoch 2. The gap that opens up between training loss and validation WER after epoch 2 is a classic sign of overfitting. With only 4,500 training examples, the model does not take long to cycle through the available data multiple times, and at some point it starts memorizing specific recordings rather than learning transferable patterns. This is precisely why early stopping and checkpoint selection based on validation WER — rather than training loss — were important design choices in our setup. Without them, we would likely have saved a model that performs well on training data but worse on anything new.

What still limits the model. Even at 45.8% WER, there is clearly room for improvement, and understanding where errors still come from is useful for framing future work. Three issues stand out. First, Darija has no agreed-upon writing system, which means different transcribers spell the same word differently — this inconsistency introduces noise into the training labels that the model has no way to resolve on its own. Second, code-switching remains a genuine challenge: when a speaker switches mid-sentence from Darija to French, the model's Arabic-configured decoder is not well-equipped to follow. Third, regional variation across cities like Casablanca, Fes, or Oujda adds acoustic diversity that 4,500 samples simply cannot cover in full.

Taken together, these observations point toward a clear direction for improvement: more data, more diverse speakers, and a decoding setup that can handle multilingual output.

2.9. Conclusion

2.9.1. Recall of Objectives and Contributions

This work set out to address a concrete and pressing gap in speech technology: the near-total absence of reliable Automatic Speech Recognition systems for Moroccan Darija and Tarifit, two widely spoken but severely underrepresented language varieties. Rather than building a system from scratch — which would have been unrealistic given the scarcity of annotated data — we chose to work with OpenAI's Whisper, a large pre-trained multilingual ASR model, and investigate how much its performance could be improved through targeted fine-tuning in a resource-constrained environment.

Our contributions can be summarized along four main lines. First, we conducted a systematic benchmark of all five Whisper variants on Moroccan Darija in zero-shot mode, providing a clear picture of where each model stands before any adaptation. Second, we fine-tuned Whisper Small on a curated subset of 4,500 Darija speech samples, demonstrating that meaningful gains are achievable even under hardware and data limitations. Third, for Tarifit — a language with virtually no publicly available annotated speech data — we manually constructed a dedicated corpus of 121 audio samples collected from YouTube, transcribed them in Latin script, and used them to evaluate and fine-tune Whisper Small, making this one of the first documented ASR studies specifically targeting Tarifit. Finally, we made our fine-tuned Darija model publicly available on the Hugging Face Hub and developed an interactive Gradio-based interface to allow direct comparison between the baseline and fine-tuned outputs, with the goal of making this work accessible and reproducible for future researchers.

2.9.2. Summary of Key Results and Main Conclusions

The experimental results were encouraging on both fronts, though they also revealed how much work remains to be done.

For Darija, the pre-trained Whisper Small model produced a baseline WER of 133.4% — a value that reflects not just poor transcription but active hallucination, where the model generates words entirely unrelated to the input speech. After fine-tuning on 4,500 samples, the WER dropped to 45.8%, corresponding to a relative improvement of 65.7%. The best checkpoint was obtained at epoch 2, after which overfitting on the limited training set prevented further gains. Across the five Whisper variants evaluated in zero-shot mode, WER ranged from 85.3% for Large to 101.7% for Tiny, confirming that model scale alone cannot compensate for the absence of dialect-specific training data.

For Tarifit, the challenge was even more fundamental, starting with the need to build a dataset from scratch. The baseline experiments across all Whisper variants produced very high error rates, with the best zero-shot result coming from Whisper Large-v3 at 121.67% WER. After fine-tuning Whisper Small on our manually constructed corpus, the WER improved from 114.58% to 88.75%, a relative improvement of 22.55%. While this gain is more modest than what we achieved for Darija, it is particularly meaningful given that the model was adapted using only 121 audio samples — a dataset size that most ASR research would consider too small to work with.

Taken together, these results lead to a clear conclusion: fine-tuning is not just helpful for low-resource dialects like Darija and Tarifit — it is necessary. Pre-trained multilingual models, however capable they may be on high-resource languages, do not transfer reliably to varieties that are absent or marginal in their training data. At the same time, the gains we observed demonstrate that the barrier to entry for this kind of adaptation is lower than it might seem, and that meaningful progress is possible even without large datasets or expensive infrastructure.

2.9.3. Future Work

The results of this study open up several directions that we believe are worth pursuing.

The most immediate need is more data. Both the Darija and Tarifit experiments were constrained by dataset size, and the overfitting patterns we observed suggest that the models have not yet reached their performance ceiling — they simply ran out of diverse training examples. Larger and more carefully curated corpora, ideally covering a wider range of speakers, regions, and speaking styles, would likely lead to substantially lower error rates. For Tarifit in particular, where our entire dataset consisted of only 121 samples, even a modest expansion of the corpus could produce meaningful gains.

A second direction concerns orthographic standardization. One of the persistent sources of noise in our Darija training data was the inconsistency in how words are spelled across different transcribers. Developing and adopting shared transcription guidelines for Darija — and for Tarifit in Latin script — would make future datasets more reliable and training more efficient.

Third, the code-switching phenomenon that characterizes everyday Moroccan speech remains largely unaddressed by our current setup. A model configured for Arabic output cannot be expected to handle seamless transitions into French or Amazigh. Future work could explore multilingual decoding strategies or dedicated code-switching corpora to tackle this challenge more directly.

From a technical standpoint, parameter-efficient fine-tuning methods such as LoRA or QLoRA — which update only a small fraction of the model's weights — could make adaptation even more accessible by reducing GPU memory requirements further, as demonstrated by Daouad et al. [10] for Tarifit. Combining these techniques with larger datasets and a standardized evaluation benchmark for Moroccan dialects would represent a significant step forward for ASR research in this region.

Finally, extending this work to other Amazigh varieties spoken in Morocco — such as Tachelhit or Central Tamazight — would help build a more complete picture of the challenges and opportunities in Moroccan multilingual speech recognition.

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2.12. Appendices

Appendix A — Cross-Model Benchmark Results (Darija):

The following table summarizes the zero-shot WER obtained for each Whisper variant evaluated on a 10-sample held-out subset of the Darija corpus, with the language forced to Arabic.

Model	WER (%)
Whisper Tiny	101.7
Whisper Base	95.3
Whisper Small	91.3
Whisper Medium	92.8
Whisper Large	85.3

Appendix B — Cross-Model Benchmark Results (Tarifit):

The following table summarizes the zero-shot WER obtained for each Whisper variant evaluated on the manually constructed Tarifit dataset.

Model	WER (%)
Whisper Tiny	129.58
Whisper Base	165.42
Whisper Small	152.92
Whisper Medium	132.50
Whisper Large-v3	121.67

Appendix C — Hugging Face Model Repository

The fine-tuned Darija model is publicly available at:
`https://huggingface.co/maghreb-ai-research/whisper-small-darija`